

ABORTION

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Learning Objectives

At the end of this session, you are expected to:

- ✓ Define abortion
- ✓ Discuss the aetiologies of abortion
- ✓ Identify the clinical features of abortion
- ✓ Describe the diagnosis and differential diagnosis of abortion
- ✓ Discuss types and management of abortion
- ✓ Explain different components of PAC
- ✓ Describe revised Penal Code of Ethiopia on Safe abortion

Definition

- According to the National Center for Health Statistics and the WHO, abortion is loss or termination of a pregnancy with a fetus aged **younger than 20 weeks' gestation or weighing <500 g.**
- **Abortion** is termination of pregnancy before viability; i.e., less than GA of 28 weeks from the LNMP or if GA is unknown fetal weight of less than 1000 gms

Cont...

- Spontaneous abortion is also known as loss miscarriage
- **Induced abortion** describes deliberate termination of pregnancy with surgery or medication of a live fetus that has not reached viability

Etiology of abortion

A. First trimester abortion :

➤ Fetal chromosomal abnormalities-

particularly Trisomy, Polyploidy, & monosomy

- It is the commonest cause of abortion in 50– 70 % of the first trimester abortions.
- The incidence of these abnormalities increased with the increase in the maternal age

Cont...

- **Maternal factor**
- **Infections:** genital tract infection , systemic infection with pyrexia & TORCH syndrome.
- **Endocrine disorders:** For example, poorly controlled diabetes mellitus, obesity, thyroid disorders , PCOS & Corpus luteum insufficiency
- **Uterine disorders:** Uterine anomalies , sub mucus fibroid & Asher man's syndrome
- **Behavioral Factors:** smoking , alcohol consumption

Cont...

Immunologic causes:

- Miscarriages are more common in women with **systemic lupus erythematosus**

➤ Environmental factors

- Environmental toxins including
 - bisphenol A, polychlorinated biphenyls, phthalates, and dichlorodiphenyltrichloroethane (DDT) (Krieg, 2016)
 - **Radiation or antineoplastic drugs** (Anderson, 2020).

Cont...

B. Second trimester abortion:

➤ Mechanical causes

- ✓ multiple pregnancies, hydramnios, leading to uterine overdistension, contractions, cervical dilatation and membrane rupture.
- ✓ Cervical incompetence (congenital & acquired)

Cont...

➤ Other causes

- Uterine anomalies and sub mucous fibroid
- Genital tract infection and PROM
- Systemic infections: HIV, Malaria, syphilis, Rubella
- Maternal health: Diabetes, Renal disease,
- Hypertension

Pathophysiology of spontaneous Abortion

✓ **Involves Maternal/Fetal/placental condition**

- **Disruption of blood flow** to the fetus
- And the growing fetus is compromised
- As the result, **necrosis** and inflammation of implantation area occurs
- Then the pregnancy becomes **partially or entirely detached** due to Uterine contractions and dilatation of the cervix resulting in expulsion

Diagnosis of abortion

➤ History

- Bleeding (duration, amount)
- Length of amenorrhea
- Abdominal Cramping
- History of interference and method employed
- Symptoms of infection

Cont...

➤ **Physical examination**

- Check V/S
- General systemic examination
- Abdominal
 - Uterine size (if palpable)
 - Location and severity of tenderness and rebound tenderness

Cont...

- **Pelvic examination**

- The amount of bleeding
- Cervical dilation status
- Foul -smelling discharge
- Check for cervical laceration
- Visualization of expelled conceptus

- **Bimanual examination**

- Estimate size of the uterus
- Check for any pelvic mass
- Check for tenderness in the pelvis

Diagnosis of abortion ...

Investigations

Laboratory

✓ **Ultrasound examination**

✓ **Urinalysis**

✓ **CBC**

- B/G and RH type, blood cross match if necessary
- Blood film – if febrile

- Blood – culture and sensitivity test
- Smear of cervical discharge
- VDRL, FBS

Differential diagnosis of abortion

- ❖ Ectopic pregnancy
- ❖ Molar pregnancy
- ❖ AUB other than Pregnancy
- ❖ Local causes
 - Urinary tract
 - GI tract bleeding

Methods to rule out ectopic pregnancy from abortion

1.culdosynthesis:-

- Position client in to Semi sitting position for 5 minutes
- Insert speculum
- Insert needle at the posterior part of cervix and aspirate

Result:-if you get non clotted blood it can be ectopic pregnancy

- if you get Serum it can be abortion
- if you get Pus it can be pelvic abscess

Therefore:- culdosynthesis positive is an indication of ectopic pregnancy.

Cont...

2. Arias-stella method

- It is **differentiation of abortion from ectopic pregnancy** using **endometrial tissue** and water through **sink and float** technique.
- Take endometrial tissue, then put on water;
- ✓ if it **floats** it indicates placental tissue (chorionic villae) and it shows **abortion**
- ✓ But if it **sinks** it indicates **ectopic pregnancy**

Classifications of abortion

- Spontaneous or induced - based on **onset**
- Early (<12wks) or late(>12wks)-based on **gestational age**
- Different clinical types - based on **clinical presentations**
- Safe or unsafe- based on **site of termination** of induced abortions
- Legal or illegal - based on **legality**

Clinical Classification of Spontaneous Abortion

- Threatened
- Inevitable
- Incomplete
- Complete
- Missed abortion
- Septic abortion.
- Recurrent abortion

Threatened Abortion

- This is defined as **bleeding through a closed cervical os** in the first 20 weeks of pregnancy and with a live embryo or fetus.
- In **60-80%** the pregnancy has the tendency to continue
- Even if miscarriage does not follow threatened abortion, later rates of **preterm birth and placental abruption** are slightly increased (Saraswat, 2010).

Threatened Abortion...

Clinical features of Threatened Abortion

- ✓ Bleeding with closed cervix
- ✓ No/mild abdominal pain
- ✓ Good general condition
- ✓ The uterus is usually correct size for date
- ✓ Presence of fetal heart activity

Threatened Abortion...

➤ Threatened abortion management

- ✓ Medical treatment usually not necessary.
- ✓ Bed rest
- ✓ Avoid – intercourse
- ✓ Avoid heavy Exercise/work, douching
- ✓ Monitor progress

Inevitable Abortion

✓ Inevitable abortion is heavy vaginal bleeding with no passage of products conception

➤ Clinical features

- severe lower abdominal pain
- Poor general condition
- Open cervix
- The uterus may be the correct size for date
- Cervical effacement
- Membrane is ruptured

Inevitable Abortion...

❖ Management of inevitable Abortion

- ✓ If pregnancy is less than 12 weeks,
 - plan for **evacuation** of uterine contents.
 - If evacuation not immediately possible:
 - Give ergometrine 0.2 mg IM (repeated after 15 min. if necessary)
 - or misoprostol 400 mcg by mouth (repeated once after 4 hours if necessary);
 - Arrange for evacuation as soon as possible.
- Ensure follow-up after treatment. (BEmONC, 2018).

Inevitable Abortion...

- ✓ **If pregnancy is greater than 12 weeks**
 - should be managed in a hospital setting
 - wait spontaneous expulsion of products of conception and
 - then evacuate uterus to remove any remaining products of conception
 - If necessary, infuse oxytocin 40 units in 1 L IV fluids at 40 drops/min.

Incomplete Abortion

A type of abortion in which only a part of conceptus is expelled and still reveals passage of tissue

➤ Clinical features

- cervix is open
- Abdominal cramp
- Bleeding from the vagina is usually persistent and severe.
- passage of conceptus tissue

Incomplete abortion ...

❖ Management of Less than 12 Weeks

- **If bleeding light to moderate** remove products of conception protruding through cervix using fingers or ring (or sponge) forceps

Incomplete abortion ...

- If bleeding heavy, evacuate uterus:
 - **MVA** is preferred method
 - If evacuation not immediately possible, give ergometrine 0.2 mg IM (repeated after 15 min if necessary) OR misoprostol 400 mcg orally (repeated once after 4 hours if necessary).
 - Ensure follow-up of the woman after treatment.

Incomplete abortion ...

➤ Management of greater than 12 Weeks

- Infuse oxytocin 40 units in 1 L IV fluids at 40 drops/min
- Evacuate any remaining products of conception from uterus by evacuation and curettage
- If necessary, give misoprostol 200 mcg vaginally every 4 hours until expulsion, but do not administer more than 800 mcg.
- Ensure follow up of the woman after treatment.

Complete Abortion

- ✓ **Complete abortion** is complete expulsion of conceptus tissue
- ✓ **Characterized by:**
 - Uterus well contracted and cervix closed
 - Cessation of vaginal bleeding
 - U/S showed empty uterine cavity
 - serial serum β -hCG levels drop quickly

Complete Abortion ...

➤ **Management of complete Abortion**

- Evacuation of the uterus usually not necessary
- Observe for heavy bleeding
- Ensure follow-up of woman after treatment

Missed Abortion

- ✓ **Missed abortion** describes dead products of conception that have been retained for days or weeks within a uterus with a closed cervical os.
- ✓ This usually happens in the second trimester

Missed abortion...

➤ Clinical features

- **Regression** of signs and symptoms of pregnancy
- **Retention** of conceptus within the uterus for more than two weeks
- **Closed cervix and uterine is size less** than calculated weeks of amenorrhea

Missed abortion...

➤ Treatment Options

- ✓ **Conservative treatment:** wait for spontaneous expulsion upto 4 weeks
- ✓ **Medical termination of pregnancy:** by Misoprostol (PGE1)
 - Cytotec vaginal (is the best) or oral tab. 200 µg, 2 tab/ 3 hrs/ up to 5 doses daily, which can be repeated next day if there is no response in the first day
 - Subsequent surgical evacuation is needed in cases of RPOC
- ✓ **Surgical evacuation of the uterus**
 - Indicated in 1st trimester missed abortion

Septic Abortion

- ✓ Septic abortion is Any of the abortion types complicated by infection
- **Classic clinical findings** includes
 - ✓ fever,
 - ✓ lower abdominal pain,
 - ✓ uterine tenderness, and
 - ✓ foul-smelling vaginal discharge
- *NB. Urinary tract infection should be excluded by urinalysis, and a complete blood count assesses leukocytosis*

Septic abortion...

- ✓ Most bacteria causing septic abortion are part of the normal vaginal flora
- ✓ The commonest organisms are :
 - ✓ Gram -ve : E.coli , streptococcus & staphylococcus
 - ✓ Anaerobics : Bacteroides
 - ✓ Rarely Cl. tetani , which is potentially lethal if not treated properly .

Septic Abortion...

➤ Types :

- **Mild** → the infection is confined to decidua
- **Moderate** → the infection extended to myometrium
- **Severe** → the infection extended to pelvis + Endotoxic shock + DIC

Septic Abortion...

Management:

- **Antibiotics** : Cephalosporin I.V + Metronidazole I.V
- **Surgical evacuation of uterus** → usually 12 hrs. after antibiotic therapy
- If products are retained, suction evacuation is performed

Recurrent Abortion

- Recurrent pregnancy loss (RPL) is classically defined as **three** or more consecutive pregnancy losses at <20 weeks' gestation or with a fetal weight <500 g.
- **Primary RPL** refers to multiple losses in a woman who has never delivered
- **Secondary RPL** refers to multiple pregnancy losses in a patient with a prior live birth.

Recurrent Abortion...

Cause

- ✓ chromosomal abnormalities,
- ✓ antiphospholipid antibody syndrome,
- ✓ specific endocrinopathies, and
- ✓ structural uterine abnormalities.

Recurrent Abortion...

Management

- Low doses of acetylsalicylic acid and low molecular weight heparin are the best solution in women suffering from recurrent spontaneous miscarriage
- This treatment combination reduces the rate by 54%

Recurrent abortion management ...

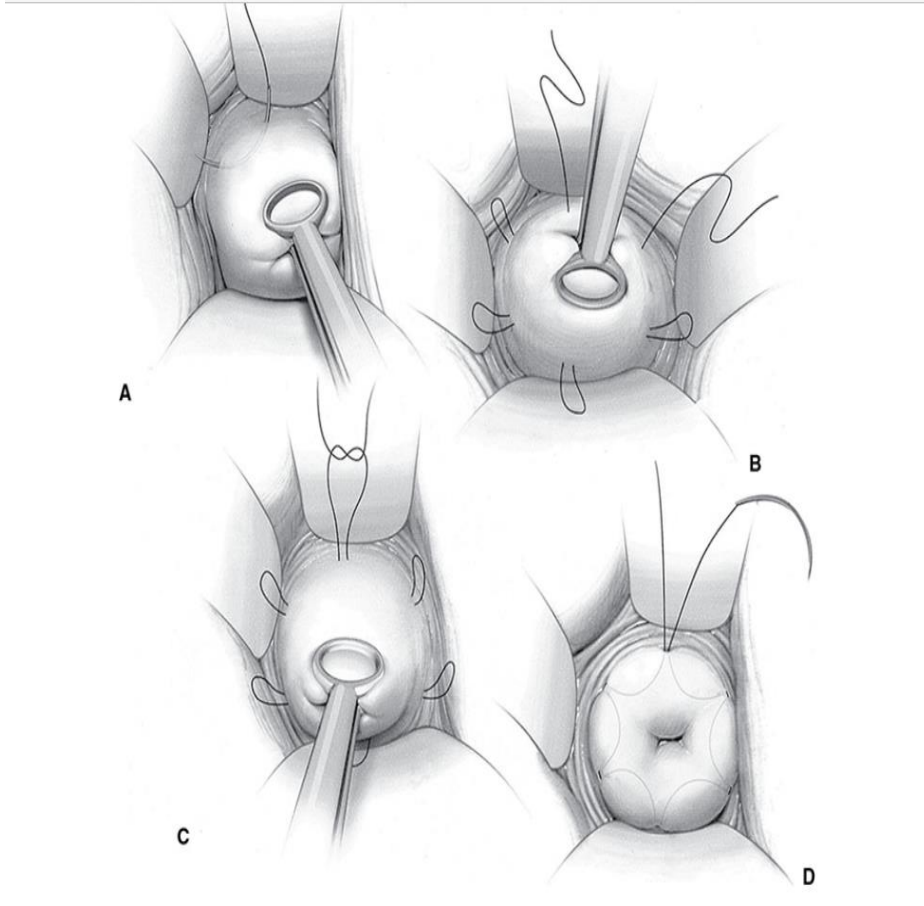


FIGURE 1 McDonald cerclage procedure for incompetent cervix.

A. Start of the cerclage procedure with a no. 2 monofilament suture being placed in the body of the cervix very near the level of the internal os.

B. Continuation of suture placement in the body of the cervix so as to encircle the os.

C. Encirclement completed.

D. The suture is tightened around the cervical canal sufficiently to reduce the diameter of the canal to 5 to 10 mm, and then the suture is tied. The effect of the suture placement on the cervical canal is apparent. A second suture placed somewhat higher may be of value if the first is not in close proximity to the internal os

Recurrent abortion management ...

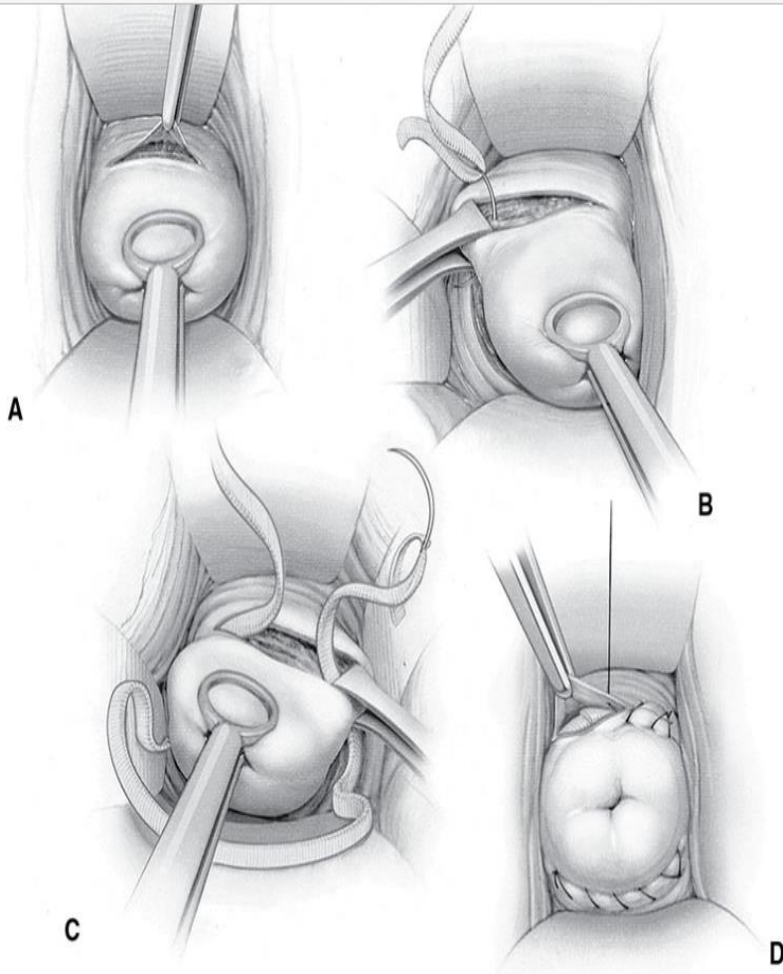


FIGURE 2 Modified Shirodkar cerclage for incompetent cervix.

- A. transverse incision is made in the mucosa overlying the anterior cervix, and the bladder is pushed cephalad.
- B. A 5-mm Mersilene tape on a swaged-on or Mayo needle is passed anterior to posterior.
- C. The tape is then directed posterior to anterior on the other side of the cervix. Allis clamps are placed so as to bunch the cervical tissue. This diminishes the distance that the needle must travel submucosally and aids tape placement.
- D. The tape is snugly tied anteriorly, after ensuring that all slack has been taken up. The cervical mucosa is then closed with continuous stitches of chromic suture.

INDUCED ABORTION

- Induced abortion is defined as termination of pregnancy with medication or surgery before fetal viability
- Induced abortion can be **safe or unsafe**
- **Safe abortion-** is defined as an abortion carried out using a method recommended by WHO, appropriate to the gestational age, and by someone with the necessary skills (WHO,2020) – the risks are very low

INDUCED ABORTION...

- Unsafe abortion the termination of unintended pregnancy either by person lacking the necessary skills or in an environment lacking the minimal medical standards or both.(WHO, 2020)
- Unsafe abortion is a global problem
- 20 - 50 % of all women who undergo unsafe abortions need hospitalization for complications.
- Unsafe abortion is, however, one of the most easily **preventable and treatable** causes of maternal death and disability.

INDUCED ABORTION...

➤ Classification of induced abortion based on purpose

- ❖ **Therapeutic abortion:** refers to pregnancy termination for medical reasons.
- The most frequent therapeutic indication currently is to prevent birth of a fetus with a significant **anatomical, metabolic, or mental deformity.**

INDUCED ABORTION...

❖ Elective abortion

- The term **elective abortion** or voluntary abortion describes the interruption of pregnancy before viability at the request of the woman, but not for medical reasons.
- Most abortions done today are elective.

Reasons for unwanted pregnancy

1. Non-use of contraception

- ✓ Most unwanted pregnancies occur in nonusers of contraceptive methods.
 - Lack of access to information about contraceptives
 - Lack of access to family planning services

Reasons for unwanted pregnancy...

2. Contraceptive failure

- ✓ Contraceptive failure results in 8-30 million pregnancies each year either from inconsistent or incorrect use of FP methods or method-related failure.

Reasons for unwanted pregnancy...

3. Sexual coercion or rape

- 20 – 50 % of women and girls report sexual abuse,
- rape or sexual coercion which carries about 5% risk of pregnancy

Reasons for unwanted pregnancy...

4. Other factors include:

- ✓ Lack of control over contraception
- ✓ Young age
- ✓ Abandonment or unstable relationship
- ✓ Mental or physical health problems, and
- ✓ Financial constraints

Grounds on which abortion is permitted in Ethiopia

Article 551 in penal code of the FDRE

Termination of pregnancy is not punishable where:

- The continuation of pregnancy puts the woman's life at risk
- Fetus has incurable impairment or deformity
- When pregnancy follows Rape or incest
- When pregnancy occurs in minor's (age <18 yrs)
- When the woman is physically or mentally unable to care for the would-be born child

Management of safe induced abortion

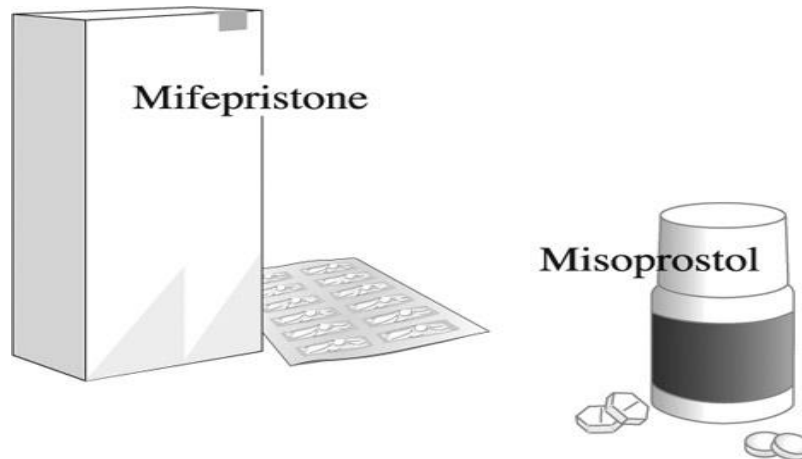
1. Medical abortion

2. Surgical abortion

- ✓ Vacuum aspiration
 - Manual and electrical
- ✓ Dilatation and evacuation

Cont...

❖ Medical abortion



- Vacuum aspiration & medication abortion are preferred methods for first-trimester abortion.
- WHO recommends replacing SC with vacuum aspiration or medication abortion.

Cont...

Mifepristone

- ✓ Is a synthetic anti-progesterone
- ✓ Leads to detachment of the pregnancy from the uterine wall
- ✓ it also dilates the cervix
- ✓ It is teratogenic if pregnancy continues after use
- ✓ Is given orally **200mg (1 tab) orally on day 1.**

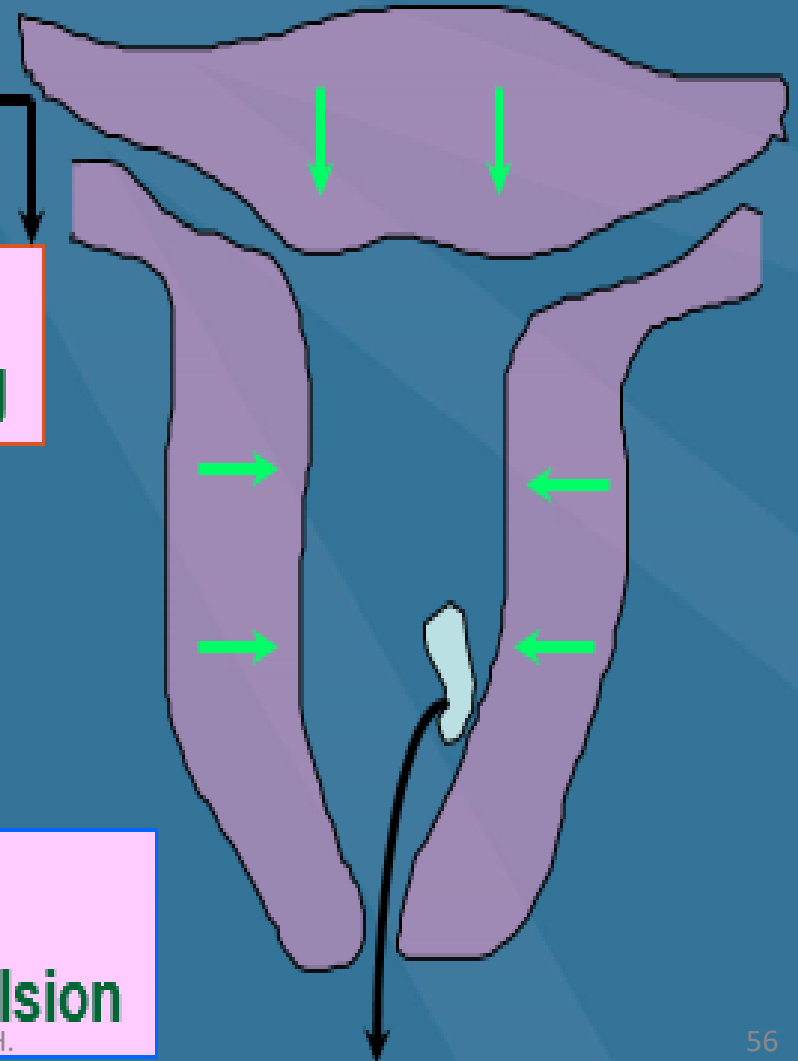
Cont...

MIFEPRISTONE
Causes
progesterone blockade

**Decidual
Necrosis**

**Cervical
Ripening**

Detachment



MISOPROSTOL

Causes uterine cramping & expulsion

Cont...

Misoprostol

- ✓ Is a prostaglandin E analogue
- ✓ Works by causing **uterine contraction and cervical dilatation**
- ✓ Is given in different route (oral, vaginal, buccal, sublingual)
- ✓ **800 (4 tabs) micro gram vaginal suppository 1-2 days after mifepristone**

Table 1. Medical management regimens for induced abortion at ≤ 24 weeks of gestation. (MOH, Ethiopia ,2021)

GESTATIONAL AGE	COMBINATION REGIMEN (RECOMMENDED ^a)		MISOPROSTOL -ONLY (ALTERNATE)	REMARK
	MIFEPRISTONE	MISOPROSTOL		
≤ 12 WEEKS	200 mg PO once day one AND	800 μ g B, PV or SL *	800 μ g B, PV or SL 24 - 48 hrs later	- Up to 9 wks ➔ Out patient 9-12 wks ➔ In patient
13 - 24 WEEKS	200 mg PO once day one AND	400 μ g B, PV or SL 24 - 48 hrs later Every 3 hrs	400 μ g B, PV or SL Every 3 hrs	Inpatient

B: buccal; PO: oral; PV: vaginal; SL: sublingual

Cont...

❖ Surgical Abortion

- surgical evacuation is performed transvaginal through an appropriately dilated cervix
- For pregnancies up to 12 weeks of gestation the preferred surgical method of termination is manual or electric vacuum aspiration.(WHO,2019)

➤ **Manual vacuum aspiration (MVA):**

❖ **Pre-procedure steps for MVA:**

- ✓ **Preoperative antibiotics:** 200 mg Doxycycline / Metronidazole 1 gm / Azithromycin 500 mg orally once within **one hour** prior to the procedure.
- ✓ Routine **cervical preparation** is not mandatory for first trimester MVA unless in special circumstances

Cont...

- ✓ **Provide pain medication** 30 minutes prior to the procedure. Diclofenac 75 mg IM stat or Ibuprofen 800 mg PO stat can be used.
- ✓ **Do bimanual examination;** assess the cervix and position of the uterus.

Steps of the MVA Procedure

1. Prepare instruments
2. Prepare the woman
3. Perform cervical antiseptic prep
4. Administer paracervical block
5. Dilate cervix
6. Insert cannula
7. Suction uterine contents
8. Inspect tissue
9. Perform any concurrent procedures
10. Process instruments

MVA Instruments



Selection of MVA cannula

Depends on uterine size and amount of dilation:

**To choose the cannula number use GA-1 and GA-2
if cx is tight.**

Uterine size (Wks)	Size of the cannula
4-6 weeks	4-7mm
7 - 9 weeks	5-10mm
10 -12weeks	8-12mm

Cont...



Signs that the uterus is empty

- ✓ Red or pink foam without tissue passing through cannula
- ✓ Gritty sensation over surface of uterus
- ✓ Uterus contracting around cannula
- ✓ Increased uterine cramping

Dilation and Evacuation (D&E)

- D & E is a surgical method which involves dilatation of the cervix and evacuation of the content of the uterus using specialized forcep
- It needs highly trained /skilled provider.
- It is performed at or beyond GA of 13 weeks

Abortion complications

Acute complications

- Incomplete abortion
- Sepsis
- Hemorrhage
- Uterine Perforation
- Bowel injury

Chronic complications

- Chronic pelvic pain
- PID
- Tubal blockage & 2^o infertility
- Ectopic pregnancy
- Increased risk of spontaneous abortion

Impact of abortion on the public health system

- **Treatment of abortion-related complications**
 - ✓ Often require several days of hospitalization
 - ✓ Blood transfusions, Antibiotics, Pain medications and other drugs.
 - ✓ More than 50 % of some hospital budgets in developing countries are used to treat complications of unsafe abortion

Key elements of post abortion care (PAC)

Five elements

1. Emergency treatment of incomplete abortion and its complications

- Treat incomplete and unsafe abortion and potentially life threatening complications

2. Counseling-about procedure, post procedure complication prevention, when to seek care etc

- Identify and respond to women's emotional and physical healthy needs and other concerns

Cont...

3. family planning services

- Help women prevent unwanted pregnancy or practice birth spacing

4. Create link with Reproductive and other health services

- Preferably provide on - site, or via referrals to other accessible facilities in provides' networks

5. Create link with Community and service provider partnerships

- Prevent unwanted pregnancies and unsafe abortion.
- Mobilize resources to help women receive appropriate and timely care for complications from abortion.

Prevention of unwanted pregnancies & unsafe abortions

- ✓ Ensure universal access to family planning
- ✓ Increase the availability of safe abortion services
- ✓ Improve the quality and accessibility of post abortion care
- ✓ Educate communities about RH and unsafe abortion

Thank you!